SUBMISSION LINKS

GitHub repository: NOT PUBLISHED - replace with the public repository URL before submission.

Live application: NOT DEPLOYED - add the public demo URL after hosting.

VISITFLOW

AI lead response & site-visit booking

Technical Design Document | Version 1.0 | 2 November 2026

PRODUCT INTENT Turn a paid property enquiry into a useful conversation and a confirmed site visit - with grounded AI, human ownership and race-safe booking.

Executive summary

VisitFlow is a focused web application for the Ahmedabad residential developer described in the selected brief. It gives buyers a mobile-first project enquiry and booking journey, while sales executives receive a shared lead inbox with source attribution, conversation history, AI summaries and follow-up ownership.

The build deliberately concentrates on the CORE BUILD: lead capture with duplicate reuse, fact-sheet-grounded AI chat, a backend slot-availability action, site-visit reservation with concurrency protection, simulated reminders, human handoff and source analytics. Project fact sheets and visit inventory are editable by an admin-capable sales account.

What is delivered

Area	Implemented behaviour
Buyer journey	Demo OTP sign-in, consent capture, project selection, conversation, human handoff and site-visit booking.
Sales workspace	Lead inbox, filters, buyer details, full chat history, owner reassignment, status and AI summary.
Automation	Gemini structured response, live slot lookup action, 24h/2h reminder logs and opt-out suppression.
Admin and insight	Project fact-sheet editor, slot create/deactivate, source metrics and response / visit indicators.

IMPORTANT RELEASE STATUS The source is prepared locally. A public GitHub URL and live app URL are intentionally not fabricated; both remain publishing steps before the reviewer submission.

1. Scope and assumptions

1.1 Selected scope

Selected client brief: Problem 2 - real estate developer AI lead-response and site-visit booking assistant. The buyer-facing part is optimised for a phone. The sales/admin side is a responsive desktop-first workspace that remains usable on smaller screens.

Included in the seven-screen target	Not built in this submission
1. Public project landing 2. Buyer chat / enquiry 3. Slot selection / confirmation 4. Sales overview and inbox 5. Lead detail / handoff 6. Source analytics 7. Admin fact sheets and slots	Full CRM replacement; portal / ad integrations; WhatsApp Business API; voice calling; payments; booking payments; live property inventory; final price quotations; real map pin; property allotment; outbound campaigns.

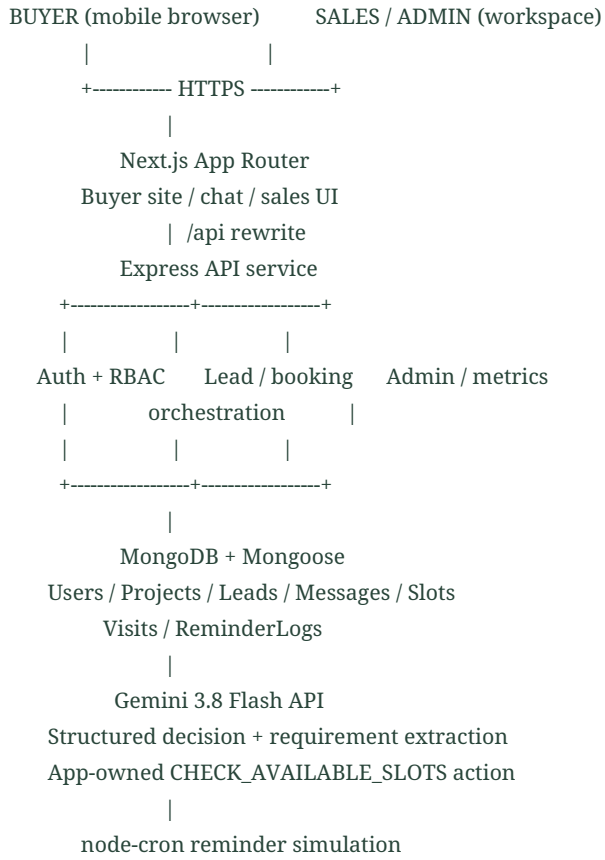
1.2 Explicit assumptions

ID	Assumption / treatment
A1	The supplied Project 1 and Project 2 labels are retained because no actual project names were provided.
A2	420 apartments and INR 65 lakh-1.2 crore are portfolio-wide facts. They are not stated as per-project totals or unit quotations.
A3	Only 2 BHK and 3 BHK configurations are known. Amenities, exact prices, floor plans, possession dates, construction status, addresses and RERA registration numbers are unknown and are not invented.
A4	Sample buyer names and lead IDs come from the brief. Phone values are synthetic placeholders; all contact channels are mocked.
A5	A buyer enquiry is keyed by normalised phone plus project. A repeated enquiry reuses the record and adds its source tag.
A6	New leads are assigned round-robin to active sales users. Admin is a permission flag on a sales user, not a third user role.
A7	Demo visit slots have capacity one by default and use Asia/Kolkata time. Seed slots are generated at 11:00, 14:00 and 16:00 for ten future days.
A8	OTP is simulated with a fixed demo code and in-memory expiry. No SMS is sent. Production requires a verified OTP provider.
A9	WhatsApp/SMS reminders are simulated as database delivery logs. No real outbound message is sent.
A10	Source-wise media spend is not supplied. Cost per visit therefore remains unavailable rather than fabricated.
A11	A past sample visit marked “Booked” remains unresolved; attendance is not inferred. Only records marked VISITED or NO_SHOW enter the show-up denominator.

2. Architecture and technology choices

The system is a small monorepo with independently runnable Next.js and Express workspaces. The browser never decides whether a slot is free and never calls the Gemini API directly.

2.1 Logical architecture



2.2 Component responsibilities

Component	Responsibility / rationale
Next.js 16 + React 19	App Router pages, responsive public and staff views, same-origin /api rewrite. SEO and rich CMS are not core to this slice.
Express 5	Central authority for authentication, access checks, lead writes, AI orchestration, booking, reminder processing and analytics.
MongoDB Atlas + Mongoose	Flexible document model for conversation records and project fact sheets; compound indexes enforce uniqueness.
JWT in HTTP-only cookie	Session token is not exposed to browser JavaScript. Secure cookie flag is enabled in production.
Gemini 3.8 Flash	Server-side structured output for message drafting, language, intent and fields. The server validates the response before acting.
node-cron	Simple single-instance reminder worker for the demo. A distributed queue/lock is required before horizontally scaling.
Vitest + Supertest + mongodb-memory-server	Business-rule and persistence tests; a separate dependency-free smoke command checks phone and AI safety rules.

Model identifier is configurable through GEMINI_MODEL; structured output is validated on the server before use.

3. Data model

The data model is intentionally small. A buyer can have one enquiry per project and multiple visit records over time. Messages are separate documents so chat history can be paged and retained independently.

Collection	Key fields	Indexes / rules
User	name, email, phone, role, isAdmin, passwordHash, active	Sparse unique email and phone. Roles are buyer or sales; isAdmin gates fact-sheet/slot controls.
Project	code, name, city, configurations, priceRangeText, description, registrationNumber, verifiedAddress, published	Unique code. Empty RERA/address fields remain unknown until admin verification.
Lead	leadCode, name, phone, normalisedPhone, project, source, sourceTags, wants, budget, timeline, language, status, assignedTo, consent, optOut, aiSummary	Unique compound index: normalisedPhone + project. Indexed status, project, assignedTo and phone.
Message	lead, sender, body, language, metadata	Indexed lead + createdAt ordering. Message body capped at 4,000 characters.
VisitSlot	project, executive, startAt, endAt, capacity, bookedCount, active, locationLabel	Unique project + startAt. Reservation uses atomic capacity predicate.
SiteVisit	lead, slot, project, status, confirmationCode	Unique lead + slot and unique confirmationCode. Status tracks CONFIRMED / CANCELLED / VISITED / NO_SHOW.
ReminderLog	visit, type, channel, status, detail, sentAt	Unique visit + type ensures one log per reminder window.

3.1 Lead lifecycle

Status	Meaning
New	Lead created; no conversation is stored yet.
In conversation	Buyer has exchanged a message with VisitFlow.
Handoff Required	AI is unsure or buyer requests a person; named owner and summary are stored.
Visit booked	A site visit has been confirmed against a slot.
Visited	Sales executive marks the buyer as attended.
Closed	Sales team closes the enquiry.

PII BOUNDARY No real buyer phone numbers are present in the seeded data. Phone fields are synthetic placeholders only. Production requires verified consent, a retention/deletion policy and appropriate data processor controls.

4. Main flows and booking correctness

4.1 Buyer enquiry and duplicate reuse

1. Buyer signs in with the demo OTP flow and submits name, project, configuration, language and contact consent.
2. Express normalises the phone number and performs an upsert on (normalisedPhone, project).
3. If the record exists, the same lead is reused and the new source is added to sourceTags; otherwise a new lead is assigned to an active executive.
4. Conversation messages are persisted against the lead. Buyer routes verify that the session phone matches the lead phone.

4.2 AI slot lookup and booking

The model may request CHECK_AVAILABLE_SLOTS. That action only reads free slots from MongoDB; it does not reserve one. The browser displays candidates, and the buyer explicitly chooses a time. Reservation is a separate authenticated API call.

5. Check that the lead exists and has recorded enquiry consent.
6. Atomically update the selected slot only when it is active, in the future and bookedCount is lower than capacity. The update increments bookedCount in the same database operation.
7. Create SiteVisit with a unique confirmation code. A unique (lead, slot) index prevents a buyer from creating duplicate bookings for the same time.
8. If SiteVisit creation or lead update fails, delete any partial visit and decrement bookedCount as compensation.
9. On cancellation, conditionally transition CONFIRMED to CANCELLED, then release one unit of slot capacity. If capacity cannot be released, restore CONFIRMED and return an error.

CONCURRENCY INVARIANT For capacity one, two simultaneous requests race on the same slot document. Only one atomic predicate can match bookedCount < capacity. A stale frontend view cannot override the server decision.

4.3 Important API surface

Endpoint	Purpose
POST /api/auth/otp/request + verify	Demo buyer sign-in; no SMS provider is called.
POST /api/auth/login, GET /api/auth/me	Sales session and current-user check.
POST /api/leads, GET /api/leads, GET /api/leads/:id	Create/reuse, list and inspect leads.
POST /api/chat/:leadId/messages	Persist a buyer message, call AI, execute slot lookup action, store assistant reply.
GET /api/slots?projectId=...	Read current free slots.
POST /api/bookings, POST /api/bookings/:id/cancel	Reserve / cancel with backend capacity checks.
GET /api/analytics	Source, visit, show-up and response metrics.
GET/PATCH /api/admin/fact-sheets; GET/POST/PATCH /api/admin/slots	Admin fact sheet and visit schedule management.

5. AI and automation design

5.1 Model input and output

The API sends a short recent conversation window, known lead fields and the selected project's approved fact sheet to Gemini. A JSON schema constrains the response to: reply, language, intent, extractedFields, requestedAction and needsHuman. Zod validates the returned object before the application uses any field.

Model field	Allowed values / use
language	English, Hindi or Gujarati; used for the response and lead preference.
intent	property_question, collect_requirements, visit_interest, human_help or other.
extractedFields	2 BHK / 3 BHK / Not sure, optional numeric budget bounds and a short timeline.
requestedAction	NONE, CHECK_AVAILABLE_SLOTS or HANDOFF. Only the server can execute the permitted action.
needsHuman	Creates a counsellor/sales follow-up task; never triggers a fee, price or outbound-contact action.

5.2 Grounding and refusal behaviour

- Only the saved Project fact sheet is supplied as factual context. Missing amenities, exact unit price, possession date, address, RERA number and discounts are not guessed.
- The portfolio-wide INR 65 lakh-1.2 crore range is allowed only as an indicative range, never as a quote for a project, configuration or buyer.
- A second server-side guard rejects monetary claims outside the portfolio range, numeric discounts and guarantee/assured claims. Unsafe output is replaced with a human-confirmation fallback.
- If the key is missing, the provider errors or the model output fails schema validation, the app returns a clear fallback and marks the enquiry for human review.
- If the buyer requests a person, or the model sets HANDOFF / needsHuman, the lead status becomes Handoff Required, an active executive is the owner, and a summary is persisted beside the original transcript.

5.3 Slot action boundary

The model cannot write SiteVisit records. CHECK_AVAILABLE_SLOTS calls the backend availability query and returns up to five live free slots. Only the buyer booking endpoint can reserve the selected slot. The server rechecks availability at the moment of booking.

5.4 Reminder automation

A node-cron task checks confirmed visits each minute for the 24-hour and 2-hour reminder windows. It writes SENT_SIMULATED or SKIPPED_OPT_OUT records to ReminderLog. A unique (visit, type) index makes log creation idempotent. The adapter never contacts WhatsApp or SMS in this demo.

OPERATIONAL LIMITATION A minute-window cron worker on one persistent process is adequate for a focused demo, not a distributed production scheduler. A production system should use a durable queue/outbox, provider delivery callbacks, retry policy and a distributed lock.

6. UI and UX decisions

The product uses a restrained forest / warm-white / clay visual palette, readable system-font fallbacks and CSS-generated architectural artwork. The application does not depend on large stock-image downloads or an external design system at runtime.

Screen	Main decision	Why
1. Public landing	Two simple project cards and one strong chat CTA.	A buyer can begin with one action without navigating a full catalogue.
2. Buyer chat	Conversation is the primary surface; language, buyer consent and requirements are collected at the start.	The chat focuses on a short buyer task and can hand off whenever facts are missing.
3. Booking	Grouped free slots, clear capacity and confirmation state. Exact location withheld until verified.	No false promise: selecting a slot is not the same as confirming it.
4. Sales overview	Four metrics, source mix, handoff count and lead inbox.	Sales staff can prioritise work quickly and spot source-level gaps.
5. Lead detail	Buyer facts, verbatim messages, AI summary, owner and status.	A human gets context without treating the model summary as authoritative.
6. Analytics	Per-source leads, visits and show-ups; metric notes stay visible.	Unknown media spend is left blank rather than inventing cost per visit.
7. Admin	Fact sheet editor and visit slot inventory in one clean workspace.	Content and inventory control are kept simple for non-technical staff.

6.1 Responsive behaviour

- Buyer journey uses one column on phones, large touch targets and a short first-screen form.
- The chat transcript has a bounded scroll area; slot suggestions are directly actionable and the booking page is separate.
- Sales tables scroll horizontally instead of shrinking content to unreadable widths.
- Loading, error, empty and success states are present on the main data-driven pages.
- Project location is not mapped because the brief supplied no verified addresses or coordinates.

DESIGN CONSTRAINT Nothing in the interface claims a final unit price, guaranteed discount, verified location or amenity not present in the client data.

7. Security, validation and privacy

Control	Implementation / remaining requirement
Authentication	JWT in an HTTP-only cookie; bcrypt password hash for sales; fixed OTP only for isolated demo.
Authorisation	Buyer may only read/message/book against a lead matching the signed-in phone. Sales routes require sales role. Fact sheet and slot administration require isAdmin.
Consent and opt-out	Lead creation requires explicit contact consent. Buyer can opt out. Reminder processor skips opted-out leads; buyer-initiated booking remains available.
Input validation	Zod schemas validate request bodies; Mongoose schema validates stored values; request JSON limited to 32 KB.
Abuse controls	Helmet security headers and per-route rate limits on authentication and chat.
Secret management	No real keys are committed. GEMINI_API_KEY and JWT_SECRET are server-side environment variables only.
Data minimisation	Synthetic phone values only. No personal data should be copied from real enquiries into the demo repository.
Audit / compliance gap	There is no complete immutable audit log for record access or fact-sheet edits. Add it before real use and complete a DPDP impact / retention review.

7.1 Production gates

- Replace demo OTP with a verified OTP service; remove fixed OTP disclosure and persist consent evidence.
- Configure approved WhatsApp/SMS templates, explicit opt-in, opt-out webhook handling and delivery status callbacks.
- Validate project registration numbers and site addresses with the sales head before publishing promotional messages or map pins.
- Add field-level audit logs, retention / deletion workflows, backups, encryption review and incident-response processes.
- Add a durable distributed reminder scheduler, monitoring and alerting; test retries and failed AI calls.

8. Testing, limitations and next steps

8.1 Test coverage

Test area	Important expected result
Phone normalisation	Indian mobile and +91 formats normalise consistently; only the exact synthetic demo number bypasses mobile-prefix validation.
AI safety filter	Allowed portfolio range passes; unit price, discount and guarantee claims are rejected.
AI provider unavailable	No-key path returns a human handoff and never pretends the AI is running.
Booking concurrency	Two buyers racing for capacity one: exactly one booking succeeds and count remains one.
Consent / opt-out	No consent prevents booking; opt-out skips reminders but does not block self-initiated booking.
Slot validity	Past or full slots are rejected; a stale UI cannot reserve them.
Reminder idempotency	Repeated scheduler runs create one log per visit and reminder type.
Access control	Buyer cannot open another buyer's lead; sales/admin APIs require the appropriate role.

VERIFICATION PERFORMED IN THIS WORKSPACE Dependency-free smoke checks passed (16 assertions) and source syntax was parsed across the JavaScript/JSX files. Full Vitest/MongoMemoryServer tests and the Next.js production build could not be executed because npm registry access timed out before dependencies could be installed.

8.2 Known limitations

- OTP and reminder delivery are simulated. Gemini requires a server-side key to run live; absent key falls back safely to handoff.
- No real project address, location pin, RERA number, individual property prices or amenity catalogue exists in the supplied brief.
- Booking uses a single-document atomic capacity increment plus compensating rollback and unique indexes, not a multi-document MongoDB transaction. Production can add a replica-set transaction or durable reservation-hold workflow.
- Reminder cron is single-process and minute-window based; it can miss a window if the host sleeps or is unavailable.
- Cost-per-visit is not available without media spend by source. Analytics are based on the seeded data and defined status outcomes only.

8.3 Next steps after review

- Publish the source repository publicly, preserve the local commit history and insert its real URL above.
- Deploy the API as a persistent Node service and the Next.js frontend; configure production secrets and add the live URL above.
- Install dependencies in a network-enabled environment, run npm test, npm run smoke and npm run build, then fix any environment-specific failures.
- Connect OTP and messaging providers, import real lead sources and complete privacy / RERA operational checks before pilot use.

SUBMISSION CHECKLIST The final assignment ZIP must contain exactly this Technical Design Document PDF and deployment.md. The source repository itself is published separately and remains public during review.